Traitle-Data

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2026-01-20

```r
library("tidyverse")
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

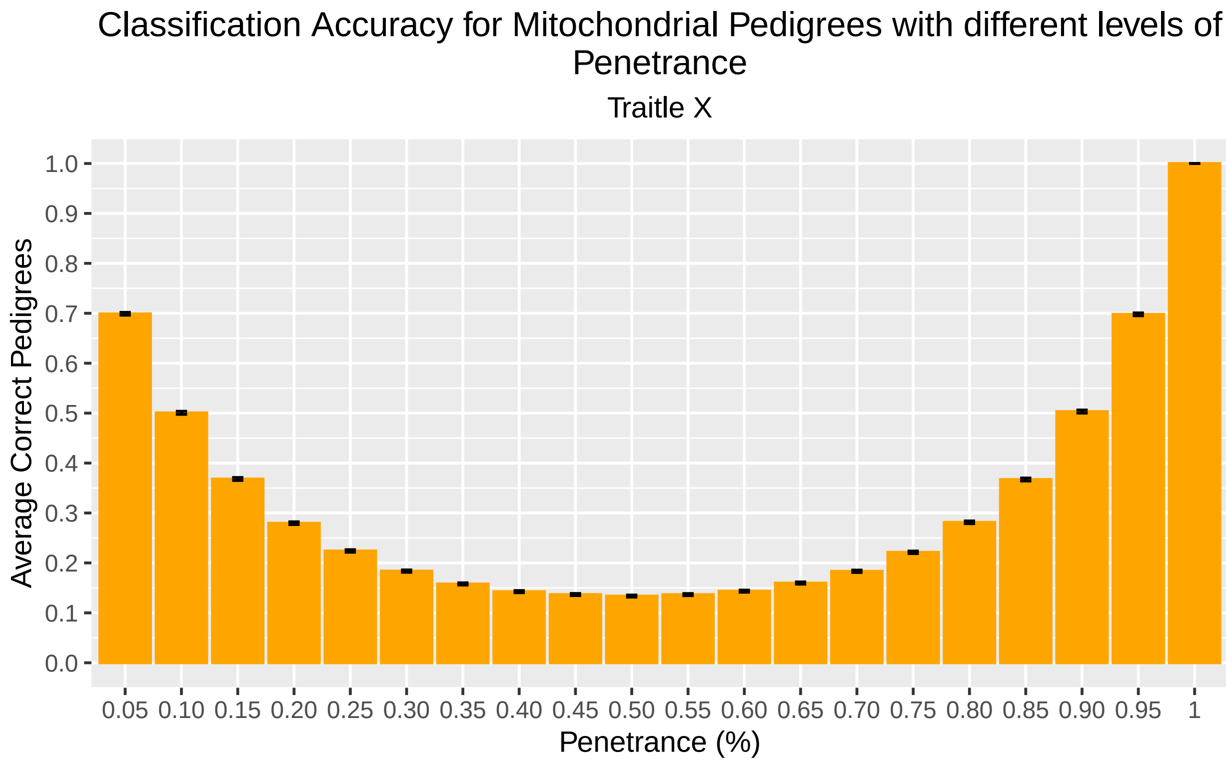
```r
library("plotrix")
library("stringr")

all.data = read.csv("results.csv")


diseaseType = all.data$DiseaseType
penetrance = all.data$Penetrance
correct = as.integer(all.data$Correct=="true")
cnt = 0
M.corrected = c()
M.sd = c()
start = 1
end = 100000
for(i in 1:20){
  M.corrected = c(M.corrected, mean(correct[start:end]))
  M.sd = c(M.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}

dfM <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60", "
ggplot(dfM, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "orange" , color = "orange") +
  labs(
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for Mitochondrial Pedigrees with different levels of Pene
```

```r
    subtitle = str_wrap("Traitle X"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each pa
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
    )+
  geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
  scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```

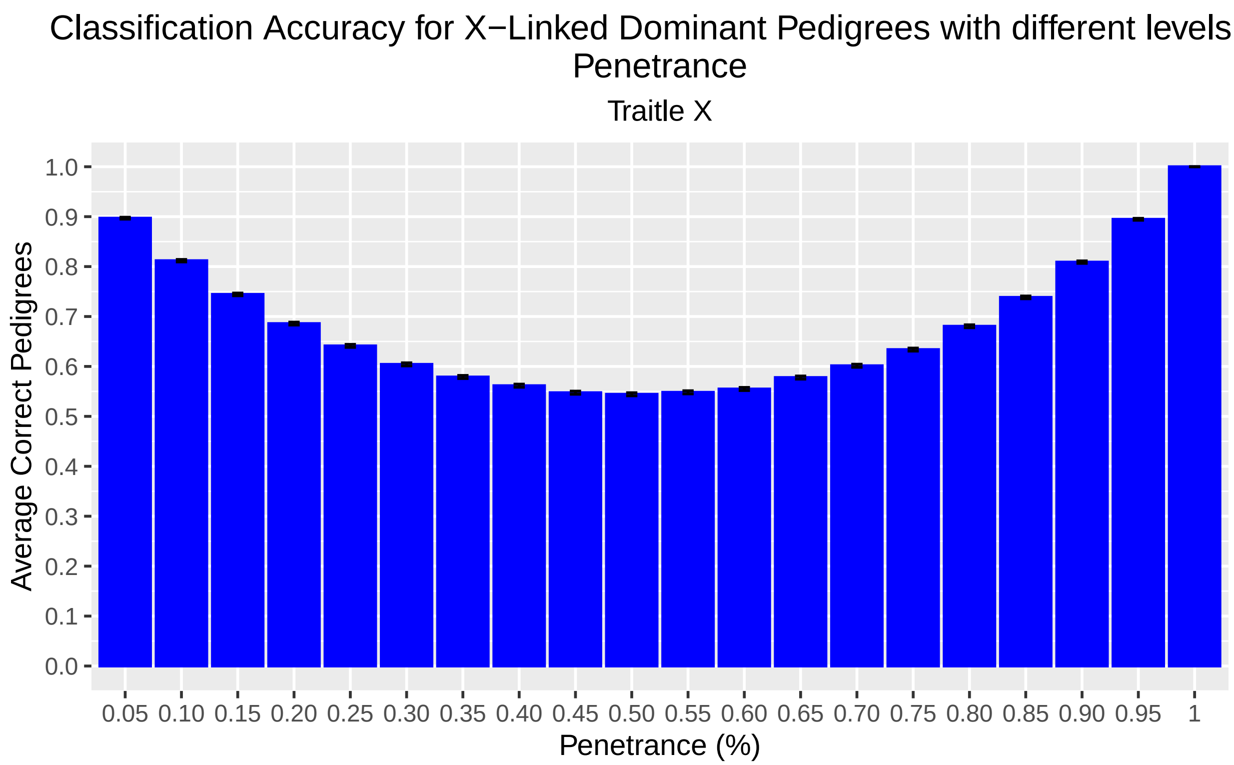


The parental generation consists of 2 individuals. In the F1 generation, each parent has 1−3 children, and in the F2 generation, each F1 individual has 2−3 children.

```r
  XLD.corrected = c()
  XLD.sd = c()
  for(i in 1:20){
    XLD.corrected = c(XLD.corrected, mean(correct[start:end]))
    XLD.sd = c(XLD.sd, 1.96*std.error(correct[start:end]))
    start = start + 100000
    end = end + 100000
  }

dfXLD <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60",
ggplot(dfXLD, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "blue" , color = "blue") +
  labs(
    x = str_wrap("Penetrance (%)"),
```

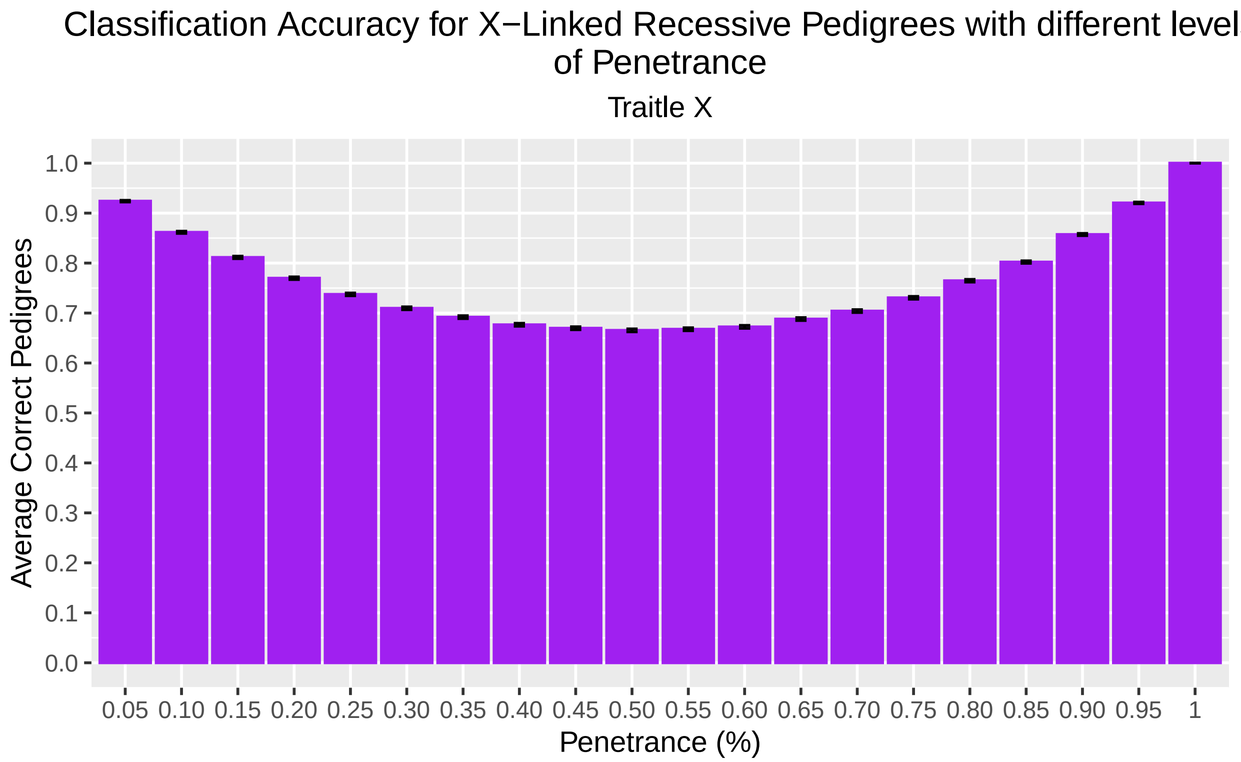
```
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for X-Linked Dominant Pedigrees with different levels of P
    subtitle = str_wrap("Traitle X"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each pa
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
    )+
  geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
  scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```



The parental generation consists of 2 individuals. In the F1 generation, each parent has 1−3 children, and in the F2 generation, each F1 individual has 2−3 children.

```
  XLR.corrected = c()
  XLR.sd = c()
  for(i in 1:20){
    XLR.corrected = c(XLR.corrected, mean(correct[start:end]))
    XLR.sd = c(XLR.sd, 1.96*std.error(correct[start:end]))
    start = start + 100000
    end = end + 100000
  }
dfXLR <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60",
ggplot(dfXLR, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "purple" , color = "purple") +
    labs(
```

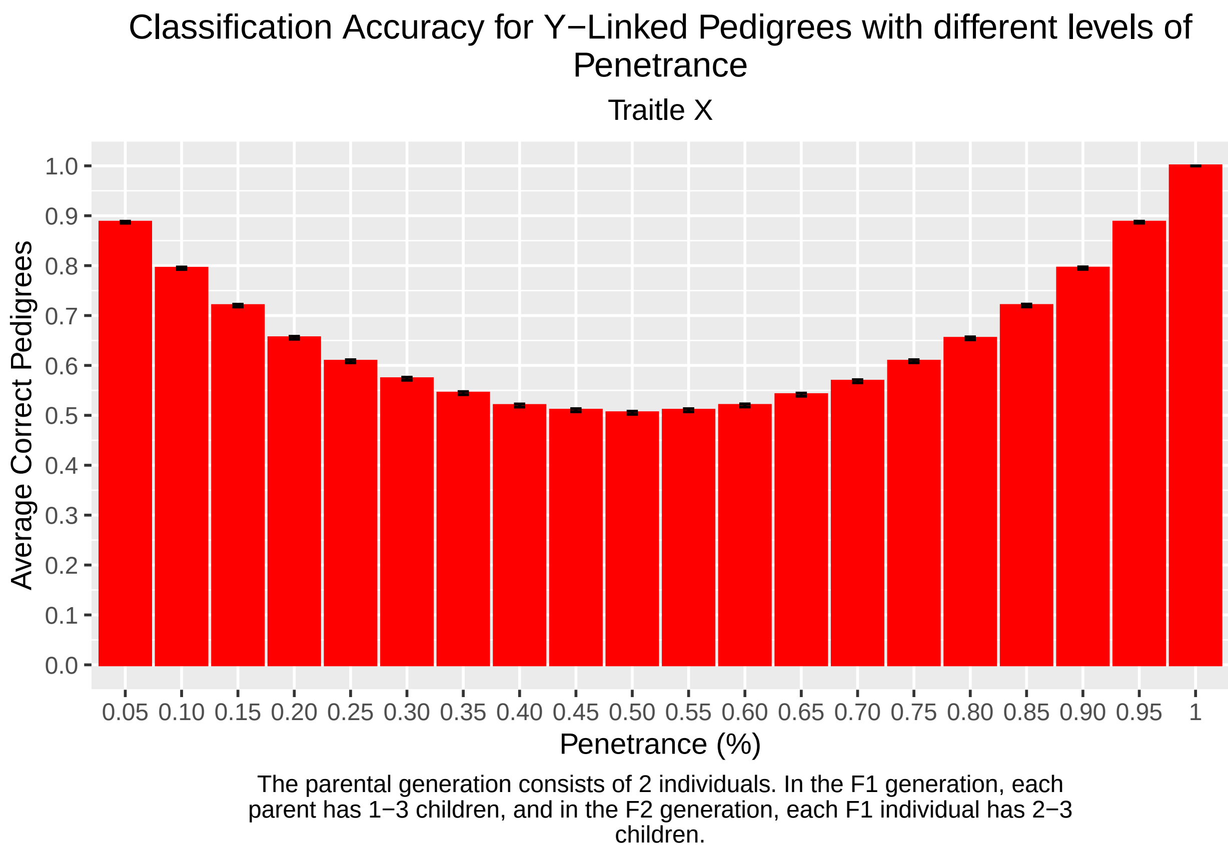
```
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for X-Linked Recessive Pedigrees with different levels of
    subtitle = str_wrap("Traitle X"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each pa
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
    )+
  geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
  scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```



The parental generation consists of 2 individuals. In the F1 generation, each parent has 1−3 children, and in the F2 generation, each F1 individual has 2−3 children.

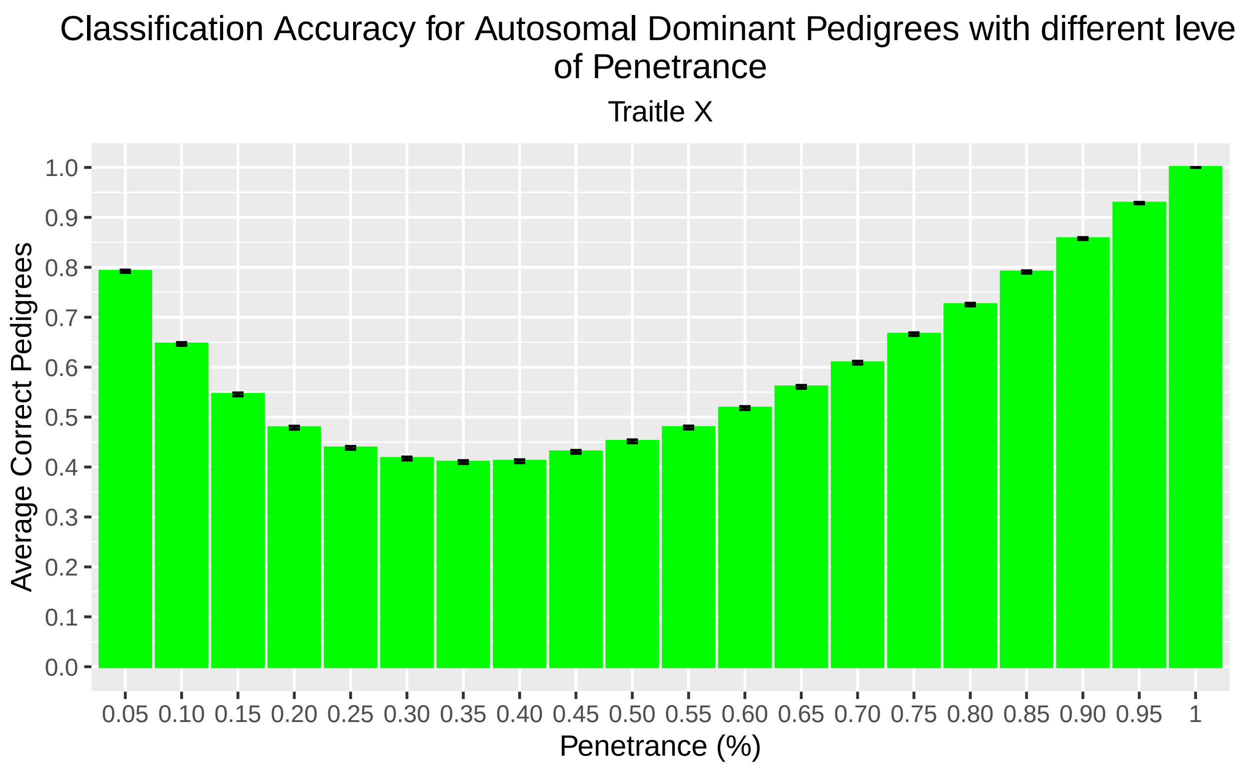
```
  YL.corrected = c()
  YL.sd = c()
  for(i in 1:20){
    YL.corrected = c(YL.corrected, mean(correct[start:end]))
    YL.sd = c(YL.sd, 1.96*std.error(correct[start:end]))
    start = start + 100000
    end = end + 100000
  }
dfYL <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60",
ggplot(dfYL, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "red", color = "red") +
```

```
    labs(
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for Y-Linked Pedigrees with different levels of Penetrance
    subtitle = str_wrap("Traitle X"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each pa
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
    )+
  geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
  scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```




```
 AD.corrected = c()
  AD.sd = c()
  for(i in 1:20){
    AD.corrected = c(AD.corrected, mean(correct[start:end]))
    AD.sd = c(AD.sd, 1.96*std.error(correct[start:end]))
    start = start + 100000
    end = end + 100000
  }
dfAD <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60",
ggplot(dfAD, aes(x = Penetrance, y = AverageCorrect)) +
```

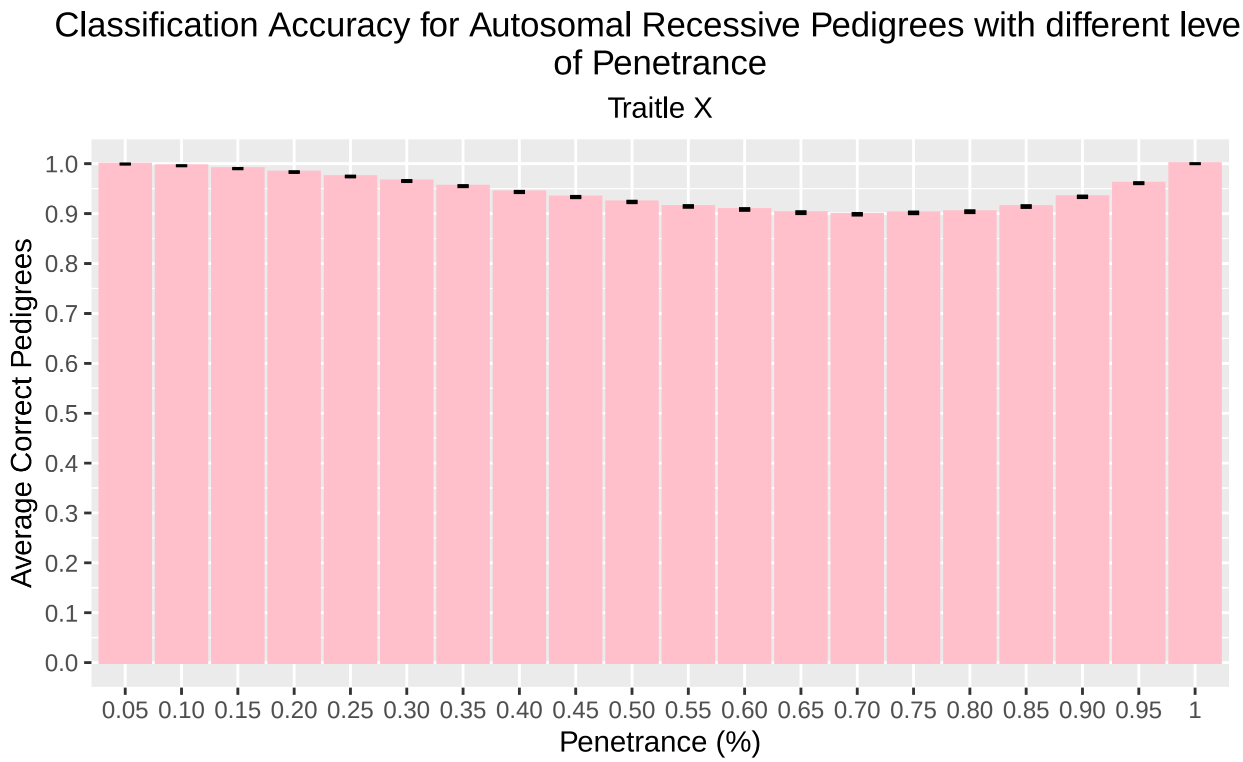
```r
geom_bar(stat = "identity", fill = "green" , color = "green") +
  labs(
  x = str_wrap("Penetrance (%)"),
  y = str_wrap("Average Correct Pedigrees"),
  title = str_wrap("Classification Accuracy for Autosomal Dominant Pedigrees with different levels of
  subtitle = str_wrap("Traitle X"),
  caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each pa
) +
theme(
  plot.title = element_text(hjust = 0.5),
  plot.subtitle = element_text(hjust = 0.5),
  plot.caption = element_text(hjust = 0.5)
  )+
geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```



The parental generation consists of 2 individuals. In the F1 generation, each parent has 1−3 children, and in the F2 generation, each F1 individual has 2−3 children.

```r
AR.corrected = c()
 AR.sd = c()
 for(i in 1:20){
   AR.corrected = c(AR.corrected, mean(correct[start:end]))
   AR.sd = c(AR.sd, 1.96*std.error(correct[start:end]))
   start = start + 100000
   end = end + 100000
 }
```

```
dfAR <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60",
ggplot(dfAR, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "pink" , color = "pink") +
    labs(
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for Autosomal Recessive Pedigrees with different levels o
    subtitle = str_wrap("Traitle X"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 generation, each p
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
    )+
  geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
  scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```



The parental generation consists of 2 individuals. In the F1 generation, each parent has 1−3 children, and in the F2 generation, each F1 individual has 2−3 children.

```
all.data$Correct = as.numeric(as.logical(all.data$Correct))

logistic.all <- glm(Correct ~ DiseaseType * (Penetrance + I(Penetrance^2)), data=all.data, family="bino
summary(logistic.all)
```

```
##
## Call:
```

```
## glm(formula = Correct ~ DiseaseType * (Penetrance + I(Penetrance^2)),
##     family = "binomial", data = all.data)
##
## Coefficients:
##                                Estimate Std. Error  z value Pr(>|z|)
## (Intercept)                    1.383358   0.005240  264.001  < 2e-16 ***
## DiseaseTypeAR                  5.321262   0.023693  224.590  < 2e-16 ***
## DiseaseTypeM                   0.052787   0.007372    7.161 8.04e-13 ***
## DiseaseTypeXLD                 1.027292   0.008084  127.085  < 2e-16 ***
## DiseaseTypeXLR                 1.297252   0.008646  150.042  < 2e-16 ***
## DiseaseTypeYL                  0.945353   0.007953  118.867  < 2e-16 ***
## Penetrance                    -8.574107   0.024681 -347.397  < 2e-16 ***
## I(Penetrance^2)               10.329219   0.025013  412.953  < 2e-16 ***
## DiseaseTypeAR:Penetrance      -5.632528   0.083046  -67.824  < 2e-16 ***
## DiseaseTypeM:Penetrance       -5.900445   0.035574 -165.863  < 2e-16 ***
## DiseaseTypeXLD:Penetrance     -0.944552   0.036228  -26.072  < 2e-16 ***
## DiseaseTypeXLR:Penetrance      0.058425   0.038268    1.527    0.127
## DiseaseTypeYL:Penetrance      -1.266106   0.035805  -35.361  < 2e-16 ***
## DiseaseTypeAR:I(Penetrance^2)   0.752696   0.069405   10.845  < 2e-16 ***
## DiseaseTypeM:I(Penetrance^2)    4.402164   0.034953  125.945  < 2e-16 ***
## DiseaseTypeXLD:I(Penetrance^2) -0.689855   0.035510  -19.427  < 2e-16 ***
## DiseaseTypeXLR:I(Penetrance^2) -1.720304   0.037357  -46.050  < 2e-16 ***
## DiseaseTypeYL:I(Penetrance^2)  -0.336046   0.035146   -9.561  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 15320802  on 11999999  degrees of freedom
## Residual deviance: 11963316  on 11999982  degrees of freedom
## AIC: 11963352
##
## Number of Fisher Scoring iterations: 7
```

```r
M.log <- all.data %>%
  filter(DiseaseType=="M")
logistic.M <- glm(Correct ~ Penetrance + I(Penetrance^2), data=M.log, family="binomial")
summary(logistic.M)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##     data = M.log)
##
## Coefficients:
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)       1.436145   0.005185   277.0   <2e-16 ***
## Penetrance      -14.474552   0.025620  -565.0   <2e-16 ***
## I(Penetrance^2)  14.731383   0.024414   603.4   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
```

```
##     Null deviance: 2525148  on 1999999  degrees of freedom
## Residual deviance: 2047673  on 1999997  degrees of freedom
## AIC: 2047679
##
## Number of Fisher Scoring iterations: 4
```

```r
XLD.log <- all.data %>%
  filter(DiseaseType=="XLD")
logistic.XLD <- glm(Correct ~ Penetrance + I(Penetrance^2), data=XLD.log, family="binomial")
summary(logistic.XLD)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##     data = XLD.log)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)      2.410650   0.006155   391.6   <2e-16 ***
## Penetrance      -9.518659   0.026521  -358.9   <2e-16 ***
## I(Penetrance^2)  9.639364   0.025205   382.4   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2499154  on 1999999  degrees of freedom
## Residual deviance: 2318304  on 1999997  degrees of freedom
## AIC: 2318310
##
## Number of Fisher Scoring iterations: 4
```

```r
XLR.log <- all.data %>%
  filter(DiseaseType=="XLR")
logistic.XLR <- glm(Correct ~ Penetrance + I(Penetrance^2), data=XLR.log, family="binomial")
summary(logistic.XLR)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##     data = XLR.log)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)      2.680610   0.006877   389.8   <2e-16 ***
## Penetrance      -8.515682   0.029244  -291.2   <2e-16 ***
## I(Penetrance^2)  8.608914   0.027746   310.3   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2175385  on 1999999  degrees of freedom
```

```
## Residual deviance: 2057393  on 1999997  degrees of freedom
## AIC: 2057399
##
## Number of Fisher Scoring iterations: 4
```

```
YL.log <- all.data %>%
  filter(DiseaseType=="YL")
logistic.YL <- glm(Correct ~ Penetrance + I(Penetrance^2), data=YL.log, family="binomial")
summary(logistic.YL)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##     data = YL.log)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)      2.328710   0.005983   389.2   <2e-16 ***
## Penetrance      -9.840213   0.025940  -379.3   <2e-16 ***
## I(Penetrance^2)  9.993173   0.024690   404.7   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2574176  on 1999999  degrees of freedom
## Residual deviance: 2370448  on 1999997  degrees of freedom
## AIC: 2370454
##
## Number of Fisher Scoring iterations: 4
```

```
AD.log <- all.data %>%
  filter(DiseaseType=="AD")
logistic.AD <- glm(Correct ~ Penetrance + I(Penetrance^2), data=AD.log, family="binomial")
summary(logistic.AD)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##     data = AD.log)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.38336    0.00524   264.0   <2e-16 ***
## Penetrance      -8.57411    0.02468  -347.4   <2e-16 ***
## I(Penetrance^2) 10.32922    0.02501   413.0   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2678920  on 1999999  degrees of freedom
## Residual deviance: 2371757  on 1999997  degrees of freedom
```

```
## AIC: 2371763
##
## Number of Fisher Scoring iterations: 4
```

```r
AR.log <- all.data %>%
  filter(DiseaseType=="AR")
logistic.AR <- glm(Correct ~ Penetrance + I(Penetrance^2), data=AR.log, family="binomial")
summary(logistic.AR)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##     data = AR.log)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)       6.70462    0.02311   290.2   <2e-16 ***
## Penetrance      -14.20663    0.07929  -179.2   <2e-16 ***
## I(Penetrance^2)  11.08191    0.06474   171.2   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 852145  on 1999999  degrees of freedom
## Residual deviance: 797741  on 1999997  degrees of freedom
## AIC: 797747
##
## Number of Fisher Scoring iterations: 7
```